

Subject: Notes on Hanson-Ibanez Paper

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Hello, here are the transcribed notes from my journal on our paper.

- 1. Introduction

- Possibly mention Ice as a dielectric medium
- Possibly mention reduced human activity time
- + Are our events during active/inactive human hours
- "HHH2022 [14]" Extra H a typo?

- 2. Units, Definitions and Conventions

- Equation (3) τ tau introduced rather suddenly. *Should there be an explanation into convolution-speak? Though it would not make much sense since this paper, analytical, does not need the mathematical proof*
 - Equation (4) $\Re$?
 - + What about adding a parameter table to section 2?*
- * note from section 4

- 3. The Askaryan Radio Array

- A5 description; split into two sentences, does not read well
- Paragraph 2 opens with a repeat sentence from just one sentence beforehand. "recorded" was replaced with "detected"
- Good on disclaiming questions on geomagnetic events.

- 4. Reconstruction Analysis

- Good on assumptions
- Calibration pulser section easy to follow
- + What about adding a parameter table to section 2?
- Rewrite "The CSWs were found in most cases to contain a reflected pulse." the sentence stands out too much in this form and abruptly stops the flow.
- 'The CSWs were found, in most cases, to contain a reflection pulse.' OR 'In most cases, the CSWs were found to contain a reflection pulse.'
- ◆ reflection or reflected, I can't decide which sounds better but this is not really a big concern.
- Clarify why relative amplitude (also called reflected pulse amplitude in table 3's caption) is set to 0, in the second paragraph after equation 8. Or rather than using two names, reduce it only to 'reflected pulse amplitude'.

- 5. Conclusion

- Insert page break for paragraph to keep it organized. Switch the description of fig 7 to come after fig 7. Currently it is both Before fig 6. and between fig 6. and fig 7.
- *It is my recommendation that the two pages of graphs be left only with their captions rather than mixed in.
- Loved the mathematical agreement with HH26 & [4]. Is there another equation from a non-whittier paper we may also demonstrate?
- Next steps: no mention of uses/comparisons with/to geomagnetic event data.

--D-I.R.

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